

Lectures: MW 2:40pm-3:55pm

Location: 227 Mudd

Instructor: Soren Galatius (sg4379@columbia.edu (<mailto:sg4379@columbia.edu>)). **Office hours:** Wednesdays 4-6pm in my office (616 Math)

TA: Charlotte Máthé (acm2311@columbia.edu (<mailto:acm2311@columbia.edu>)). **Office hours:** Tuesdays 5-8pm in the math help room (406 Math)

Textbook

No official textbook. We will loosely follow the following references

[GW] a set of [notes](https://courseworks2.columbia.edu/courses/236502/files/26378908?wrap=1) (<https://courseworks2.columbia.edu/courses/236502/files/26378908?wrap=1>) that I wrote with Nathalie Wahl for a similar course in Copenhagen.

[K] a set of [notes](https://courseworks2.columbia.edu/courses/236502/files/26378911?wrap=1) (<https://courseworks2.columbia.edu/courses/236502/files/26378911?wrap=1>) by Alexander Kupers for a similar course in Toronto.

Other presentations (auxiliary texts):

[GH] [Greenberg and Harper](https://www.amazon.com/Algebraic-Topology-Course-Mathematics-Lecture/dp/0805335579) [↗](https://www.amazon.com/Algebraic-Topology-Course-Mathematics-Lecture/dp/0805335579) (<https://www.amazon.com/Algebraic-Topology-Course-Mathematics-Lecture/dp/0805335579>): Algebraic Topology, a first course. Similar in style to [GW].

[H] [Hatcher's book](https://pi.math.cornell.edu/~hatcher/AT/ATpage.html) [↗](https://pi.math.cornell.edu/~hatcher/AT/ATpage.html) (<https://pi.math.cornell.edu/~hatcher/AT/ATpage.html>) is available in print and as pdf for download.

[M] [May's book](http://www.math.uchicago.edu/~may/CONCISE/ConciseRevised.pdf) [↗](http://www.math.uchicago.edu/~may/CONCISE/ConciseRevised.pdf) (<http://www.math.uchicago.edu/~may/CONCISE/ConciseRevised.pdf>) is available in print and as pdf for download.

Prerequisites

MATH UN2010 (Linear algebra), MATH GU4041 (Introduction to modern algebra I), and MATH GU4051 (Topology).

Math GU4052 (Introduction to modern algebra II) is recommended as a pre- or co-requisite.

Experience with proofs will be assumed.

Course Description and goals

This course is an introduction to the study of topological spaces using algebraic invariants. We will begin with the notion of homotopy equivalence. The first part of the course will introduce the fundamental group and how to compute it via covering spaces.

The second part of the course will introduce homology, its properties, and its axiomatic characterization. The last part of the course will introduce cohomology and the cup product.

Homework

12 assignments, due weekly. Each homework counts towards 2% of the final course grade, up to a maximum of 10%.

Assignments will be posted here as the course progresses. Solutions must be uploaded to Gradescope by the deadline.

You're encouraged to work with others, ask questions, and seek help, but all submitted work must be your own and reflect your personal understanding of the material.

Grades and Exams

Two in-class midterms and one in-class final exam (tentatively 1:10-4:00pm on May 13). Midterms and final exams are closed-book, no-notes.

Your final grade for the course will be based on a weighted average as follows: Homework: 10%, Midterms: 20% each, Final: 50%.

Course policies

Academic integrity: for school-specific policies, see: [Columbia College / SEAS \(https://www.cc-seas.columbia.edu/integrity\)](https://www.cc-seas.columbia.edu/integrity), [General Studies \(https://bulletin.columbia.edu/general-studies/academic-policies/academic-integrity-community-standards/\)](https://bulletin.columbia.edu/general-studies/academic-policies/academic-integrity-community-standards/), [Barnard College ↗ \(https://barnard.edu/students-roles-and-responsibilities\)](https://barnard.edu/students-roles-and-responsibilities).

AI policy: as mentioned above, all submitted work must be your own and reflect your personal understanding of the material. Use of AI-generated content is considered unauthorized assistance under Columbia's academic integrity policies.

Religious observances: In accordance with New York State law, no student may be penalized for absences related to religious observance. Students who expect to miss class, an exam, or an assignment due to a religious holiday must notify the instructor at the beginning of the term so that appropriate arrangements can be made. Even when absences are excused, students are responsible for keeping up with course requirements. If a conflict arises, it is the student's responsibility to work with the instructor in advance to determine how the work will be completed.

Disability accommodations: [Columbia \(CC, SEAS, GS\) \(https://health.columbia.edu/services/register-disability-services\)](https://health.columbia.edu/services/register-disability-services), [Barnard ↗ \(https://barnard.edu/disabilityservices/register-cards\)](https://barnard.edu/disabilityservices/register-cards). If you are eligible for exam-related accommodations, please be aware that advance notice is required through the Disability Services exam request webform. Late requests are not guaranteed and depend on space and staffing availability.

Calendar

Date	Summary		
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1/21	Introduction and overview. Homotopies and homotopy equivalence. Being homotopic is an equivalence relation on the set of continuous maps $X \rightarrow Y$. Homeomorphism implies homotopy equivalence. Examples (with proof) of maps that are homotopic, spaces that are homotopy equivalent. Examples (without proof) of maps that are not homotopic, and spaces that are not homotopy equivalent. Contractible spaces.	[GW §1] [GH §2-3], [H §1.1], [M §1.2].	
1/26	The fundamental group of a based space. Prerecorded video lecture due to snow storm, see here for the list of videos (https://courseworks2.columbia.edu/courses/236502/discussion_topics/1557202).	[GW §1.1-1.4]	
1/28	Proof that the group homomorphisms induced by a based map is an isomorphism, if the map is a homotopy equivalence. Statement of the path lifting and homotopy lifting theorems ([GW, theorems 2.2.1 and 2.2.2]) for a covering map $p : X \rightarrow Y$. Definition of the right action of the group $\pi_1(Y, y)$ on the set $F = p^{-1}(y)$. Proof that the action is well defined, using the path and homotopy lifting theorems. Statement that the action is transitive when X is path connected, and free when $\pi_1(X, x) = \{e\}$ for all $x \in F$.	[GW §2.1-3]	
2/2	Proof of path and homotopy lifting. Applications.	[GW, §2.2]	
2/4	Freeness and transitivity of the action of $\pi_1(Y, y)$ on $F = p^{-1}(y)$. Examples of covering maps, from covering space actions. The fundamental group of a quotient space.	[GW, §2.4-2.6]	HW1 due
2/9	Examples of the main theorem from last lecture (fundamental group of a quotient space $Y = X/G$, for a covering space action of a group G on a simply connected space X). The circle, a torus, a Klein bottle, real projective space. Introduction to the van Kampen theorem.	[GW 2.6, 3.1]	
2/11	Statement of van Kampen's theorem ([GW Theorem 3.1.1]: the fundamental group is generated by images of fundamental groups of U and V).	[GW 3.1]	HW2 due

	Introduction to free products of groups.		
2/16	More lifting theorems: for a path-connected, locally path connected space W , and a covering map $p : X \rightarrow Y$, when can a continuous map $f : W \rightarrow Y$ be written as $f = p \circ g$ for some continuous $g : W \rightarrow X$?	[GW, §2.7]	
2/18	Classification of covering maps.	[GW, §2.8]	HW3 due
2/23	Pre-recorded lecture (https://courseworks2.columbia.edu/courses/236502/discussion_topics/1575059) due to snow. Categories and functors.		
2/25	Singular homology. Introduction and main definitions.	[GW §5], [K §1.2]	HW4 due
3/2	First in-class midterm. Here is a practice exam (https://courseworks2.columbia.edu/courses/236502/files/26712111?wrap=1) .		
3/4	Proof that $\partial \circ \partial = 0$. Example of how to name elements of $H_p(X)$, Proof that $H_0(X) \cong \mathbb{Z}$ for path connected X .		HW5 due
3/9	SG out of town, lecture by Lucy Yang. Chain complexes. Homology of a chain complex. Exact sequences (long and short). A short exact sequence of chain complexes induces a long exact sequence of homology groups.	[GW §6.2-6.5, 6.7]	
3/11	Relative homology. Long exact sequence associated to a pair of spaces (X, A) . Long exact sequence associated to a triple of spaces. Excision and homotopy invariance (statement only). Relative homology $H_p(D^n, \partial D^n)$ for all p and n .	[GW §6.9-6.10]	HW6 due
3/16	Spring recess. No lecture.		
3/18	Spring recess. No lecture.		
3/23	Relative homology $H_p(D^n, \partial D^n)$ for all p and n , using excision and homotopy invariance. Homology of a sphere. Geometric consequences: there is no continuous map from a disk to its boundary, which restricts to the identity map of the	[GW §6.10-6.12]	

	boundary [GW, Theorem 6.12.1]. Brouwer's fixed point theorem: for any continuous map $f : D^n \rightarrow D^n$, there exists an $x \in D^n$ with $f(x) = x$.		
3/25	Degree of a map $S^n \rightarrow S^n$. The abelianization of a group. The relationship between $\pi_1(X, x)$ and $H_1(X)$ for path connected spaces.	[GW, Ex 4.3.5] [GW §6.1 and Definition 9.2.1]	HW7 due
3/30	Chain homotopies. Homotopy invariance. More on excision.	[GW §7.1, 7.4, 7.5]	
4/1	Second in-class midterm. Here is a practice exam (https://courseworks2.columbia.edu/courses/236502/files/27049219?wrap=1) .		
4/6	SG out of town, lecture by Lucy Yang. Proof of the excision theorem.	[GW §7.2, 7.3]	
4/8	The Mayer-Vietoris long exact sequence. Δ -complexes.	[GW §8.1, 8.3, 8.4]	HW9 due
4/13	Δ -complexes and simplicial homology. Simplicial chains and simplicial homology of the torus. Using simplicial homology to name a generator of $H_n(S^n)$, degree of a reflection map $S^n \rightarrow S^n$.	[GW §8.2, 8.3, 8.4]	
4/15	Degree of the antipodal map $S^n \rightarrow S^n$. Proof of the comparison between simplicial and singular homology for finite dimensional Δ -complexes.		HW10 due
4/20	More applications of simplicial homology. Rank of a finitely generated abelian group, behavior of rank in short exact sequences. Betti numbers. Euler characteristic. Generalized Jordan Curve Theorem statements (reduced homology of $S^n \setminus K$, when K is homeomorphic to I^ℓ or S^ℓ).		
4/22	Proof of the generalized Jordan Curve Theorem. Introduction to cohomology.	[GW §9.4] [H §3.1]	HW11 due
4/27	For a space X and an abelian group G , we defined $C^*(X; G)$, singular cochains of X with coefficients in G , by applying the functor $\mathbf{Hom}(-, G)$ to singular chains. It comes with a coboundary operator $\delta^p : C^p(X; G) \rightarrow C^{p+1}(X; G)$. Singular cohomology $H^p(X; G)$ is then defined as $\mathbf{Ker}(\delta^p)/\mathbf{Im}(\delta^{p-1})$.		

	We briefly discussed one method for calculating singular cohomology: in the presence of a Δ-complex structure, one can dualize the simplicial cochains. The cup product $\phi \cup \psi \in C^{p+q}(X; R)$ of cochains $\phi \in C^p(X; R)$ and $\psi \in C^q(X; R)$ when R is a ring.		
4/29	Several formal properties of the cup product (without proof). The Leibniz rule for interaction of cup product and coboundary, proof that there is a well defined induced cup product of cohomology classes. The natural homomorphism $H^n(X; G) \rightarrow \text{Hom}(H_n(X), G)$ and the statement of the universal coefficient theorem.		HW12 due
5/4	Some examples without proof.		
5/13	Final exam 1:10-4pm in the usual class room (Mudd 227).		